

Basic of git

Introduction to git and collaborative
working

Nikhil Nath R

What we will do

- Key concepts
- Usage: Simple
- Connecting to remote
- Branching
- Contribute to project

Key concepts



Git → Save Game

Git is a distributed version control system that tracks versions of files.

What is version control

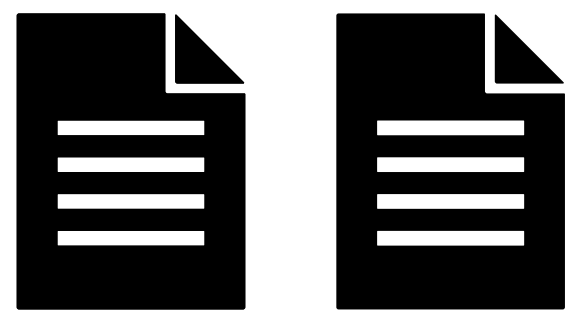
- System which keeps a track of your changes
- Allows for collaborative development
- Allows you to know who made what change and when
- Allow you to go back to any previous state

commit

- The act of creating a snapshot
- A project is made up of a bunch of commits
- Has 3 information
 1. How files changed from previous commit
 2. A hash code, uniquely identify the commit
 3. A reference to the previous commit

staging area

- Called the **staging area** or **index**
- An intermediate area where commits can be formatted and reviewed before completing the commit



Untracked changes

Review changes

Staging area

Repository

repository

- A collection of all files and all the history of the project
- Can be stored in local machine or a **remote** (GitHub, GitLab etc.)
- Often shortened to **repo**
- The act of copying from remote to machine is called **cloning**

HEAD

- HEAD is a reference to the current check-out commit in your repository
- By default it is a pointer or symbolic reference to the latest commit in your branch

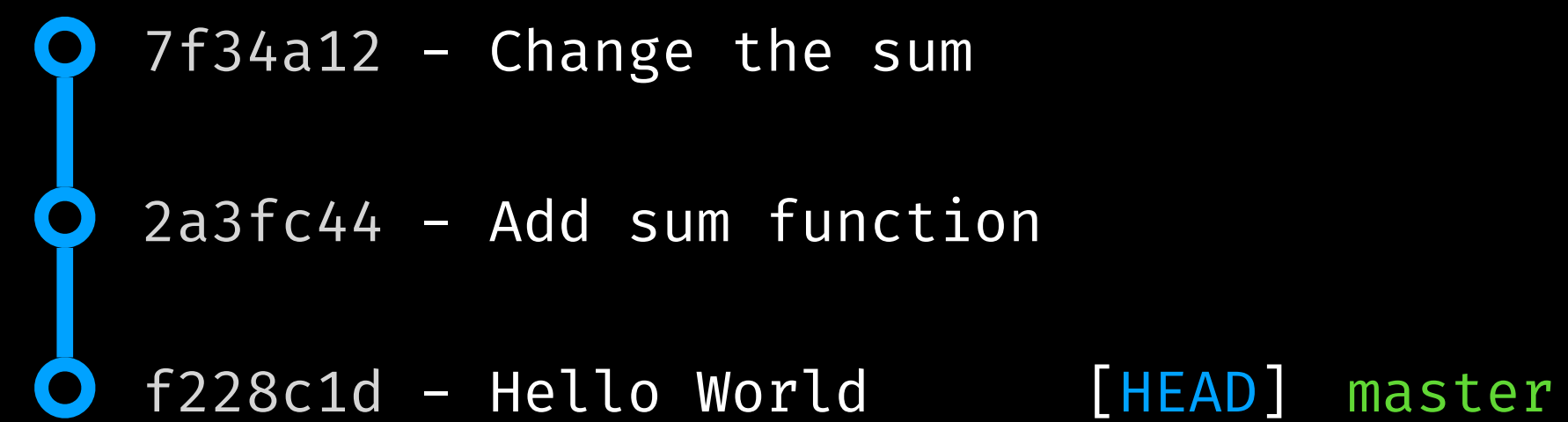
.git/

Main.java

```
package com.artusacademy

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World");
    } System.out.println("Hello World " + sum);
} }
}
```

git repository



A vertical line of three blue circles connected by a blue line, representing a sequence of commits. The circles are positioned to the left of the commit messages.

```
graph TD
    A((7f34a12 - Change the sum)) --- B((2a3fc44 - Add sum function))
    B --- C((f228c1d - Hello World))
    C --- D[HEAD master]
```

7f34a12 - Change the sum

2a3fc44 - Add sum function

f228c1d - Hello World [HEAD] master

630M

Total projects

1.12B

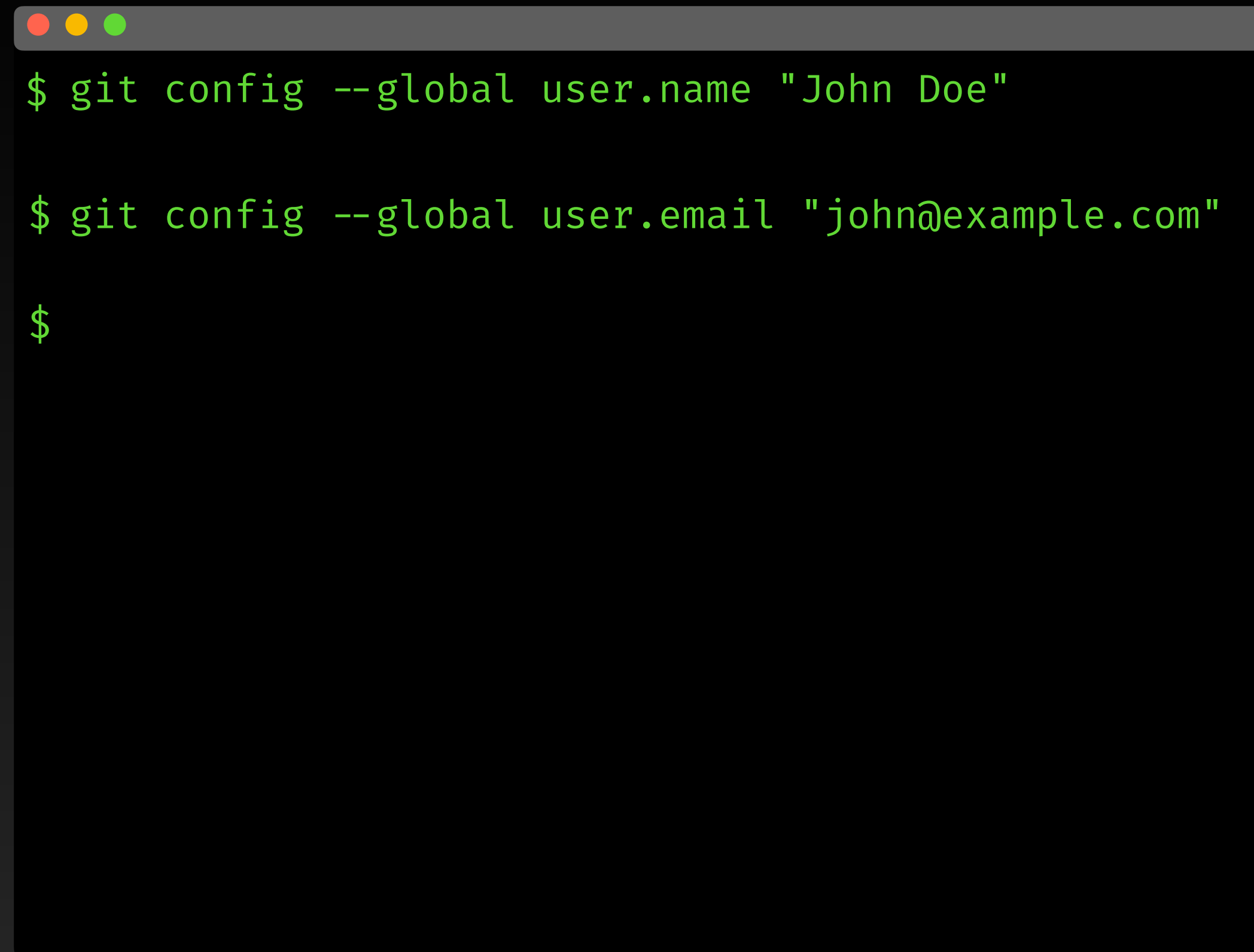
Total contribution in the year

Github Octaverse insights 2025

Usage: *Simple*

Commands

- `git init`
- `git add`
- `git commit`
- `git status`
- `git diff`
- `git log`

A terminal window with a dark gray background and a light gray title bar at the top. The title bar contains three colored window control buttons (red, yellow, green) on the left. The terminal displays three lines of green text: a command to set the global user name, a command to set the global user email, and a prompt character.

```
$ git config --global user.name "John Doe"

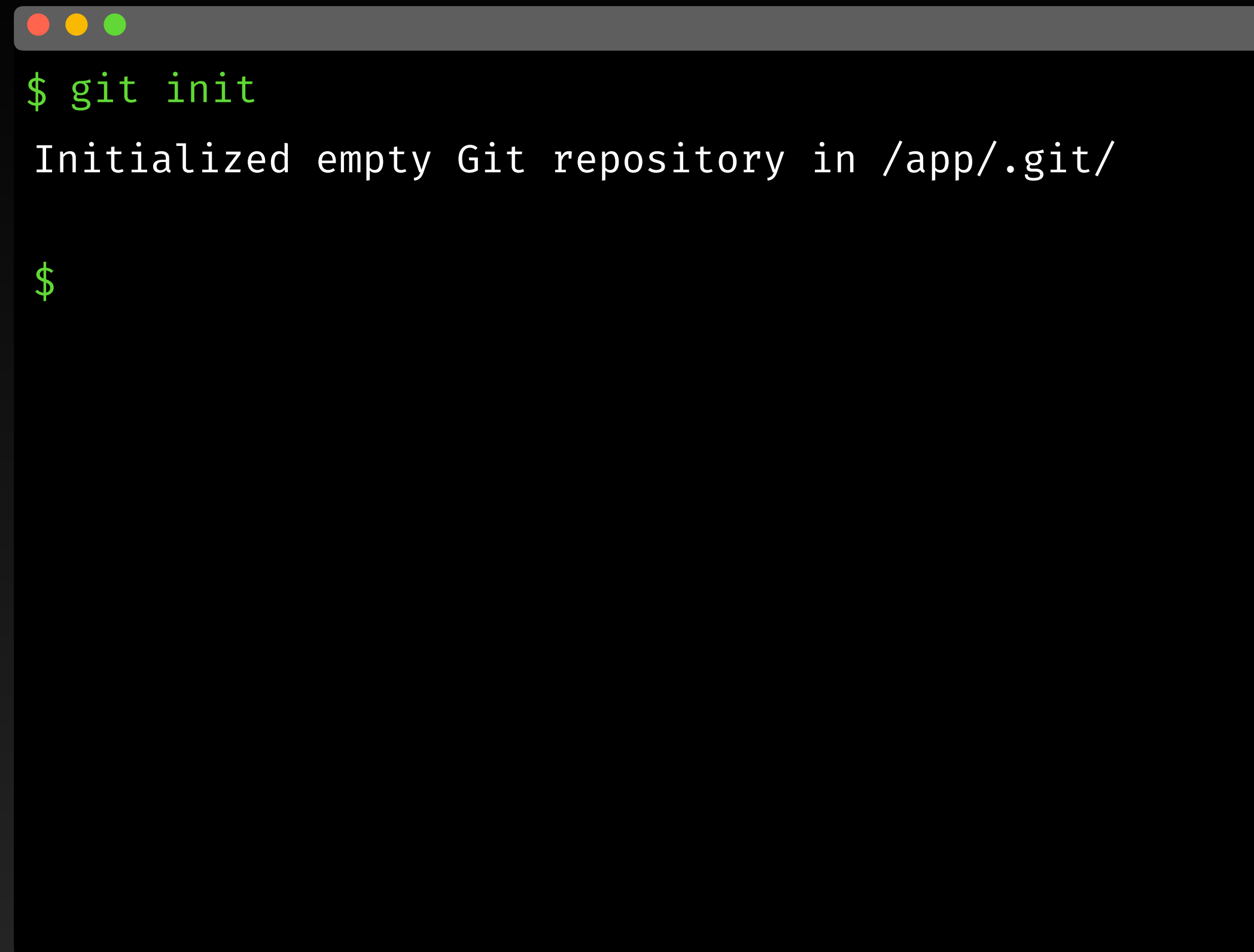
$ git config --global user.email "john@example.com"

$
```

git init

<https://git-scm.com/docs/git-init>

- A one-time command you use during the initial setup of a new **repository**.
- This will create a new master (main) **branch**.



```
$ git init
Initialized empty Git repository in /app/.git/

$
```

A terminal window with a dark background and a light gray title bar. The title bar contains three colored window control buttons (red, yellow, green) on the left. The terminal displays the command `$ git init` in green text, followed by the output `Initialized empty Git repository in /app/.git/` in white text. A new green prompt `$` is shown on the next line.

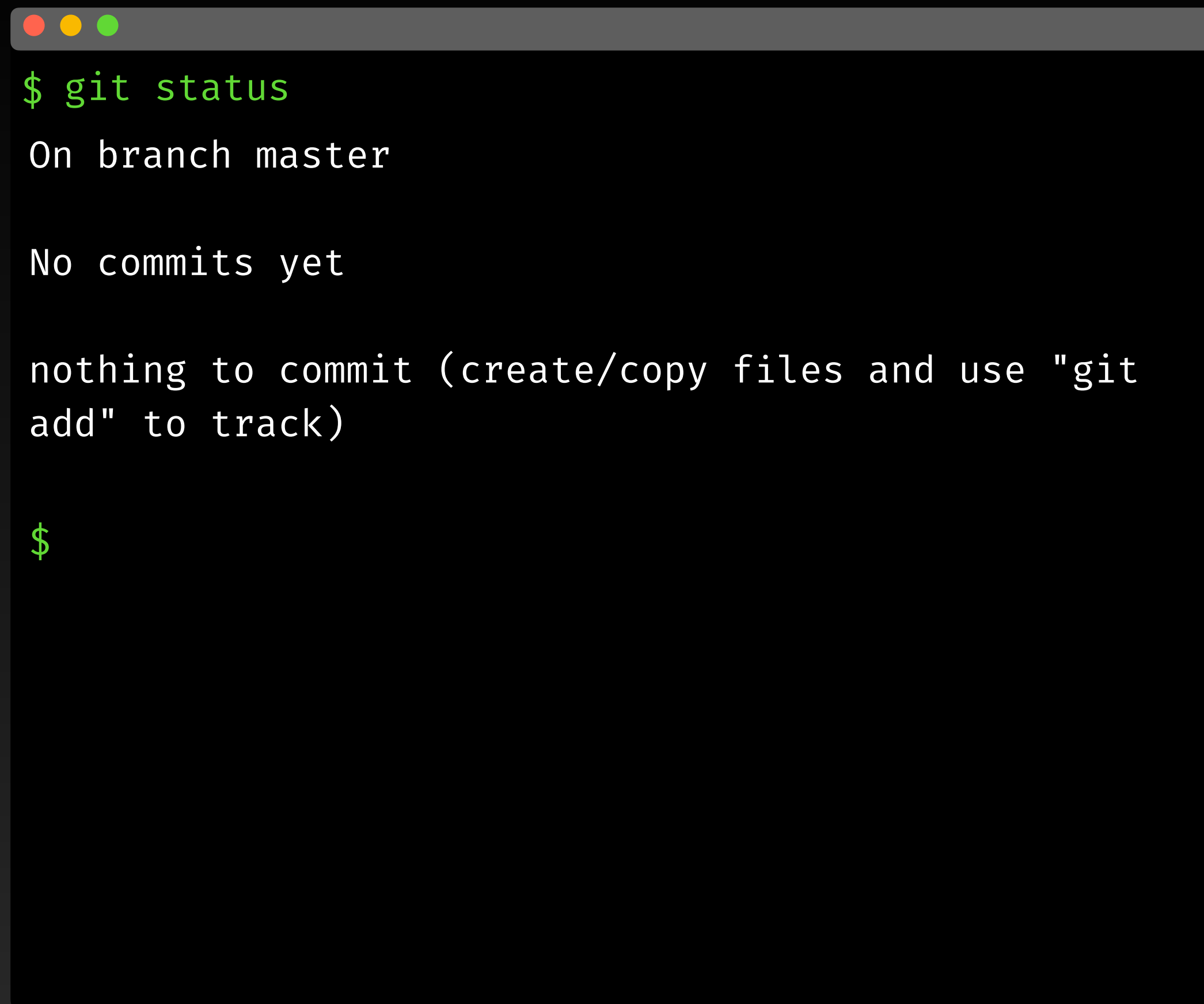
created a `.git` folder

- This is the "thing" which makes your project a Git repository
- `.git` contains all information required for version control.
- If you want to clone your repository, copying `.git` is enough.

git status

<https://git-scm.com/docs/git-status>

- Show the working tree status
- Gives information about the files which are not added to the repository.



```
$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git
add" to track)

$
```

.git/
Main.java

```
package com.artusacademy

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

```
$ git status
```

On branch master

No commits yet

nothing to commit (create/copy files and use "git
add" to track)

```
$
```

.git/
Main.java

```
package com.artusacademy

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

```
$ git status
```

On branch master

No commits yet

Untracked files:

(use "git add <file> ..." to include in what will
be committed)

Main.java

nothing added to commit but untracked files
present (use "git add" to track)

```
$
```


git add

<https://git-scm.com/docs/git-add>

- This command updates the index using the current content found in the working tree, to prepare the content staged for the next commit.
- `git add .` – add all untracked changes to stage
- `git add <path>` – add all files in given path to stage

.git/
Main.java

```
package com.artusacademy

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

```
$ git add .
$ git status

On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file> ..." to unstage)
        new file:   Main.java
```

.git/
Main.java

```
package com.artusacademy

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

```
$ git status
```

On branch master

No commits yet

Changes to be committed:

(use "git rm --cached <file> ..." to unstage)

new file: Main.java

Changes not staged for commit:

(use "git add <file> ..." to update what will be committed)

(use "git restore <file> ..." to discard changes in working directory)

modified: Main.java

git diff

<https://git-scm.com/docs/git-diff>

- Show changes between the working tree and the index or a tree
- `git diff` – show changes in untracked file
- `git diff --staged` – show changes in staging area

.git/
Main.java

```
package com.artusacademy

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

```
$ git diff
diff --git a/Main.java b/Main.java
index b53f006..02c8471 100644
--- a/Main.java
+++ b/Main.java
@@ -2,7 +2,7 @@ package com.artusacademy

    public class Main {
        public static void main(String[] args) {
-           System.out.println("Hello World");
+           System.out.println("Hello World!");
        }
    }

$
```


.git/
Main.java

```
package com.artusacademy

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

```
$ git add .
$ git status

On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file> ..." to unstage)
        new file:   Main.java
```

git commit

<https://git-scm.com/docs/git-commit>

- Create a new commit containing the current contents of the index and the given log message describing the changes
- `git commit -m "<commit message>"`

.git/
Main.java

```
package com.artusacademy

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

```
$ git commit -m "Initial commit"
[master (root-commit) f228c1d] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 Main.java

$
```


Good commit message

- Use an imperative tone like, giving an order or request
- Do not end in punctuation
- Do not exceed 50 characters for first line, be precise
- More information can be added in description and lines should not exceed 72 characters

Good commit message: Examples

- Add fix for dark mode toggle state
- Update incorrect contact number in footer
- Changed style
- oops
- Fixed a bug which prevents users from logging into the application with a phone number

git log

<https://git-scm.com/docs/git-log>

- Show the history of commits existing in the repository.
- The output is given in reverse chronological order by default.



```
$ git log
```

```
commit f228c1db183cbb261c7c187fe60e4ae5fc0228e2  
(HEAD → master)  
Author: nikhilnathr <nikhil@artusacademy.com>  
Date:   Mon Oct 21 01:02:07 2024 +0530
```

```
    Initial commit
```

```
$
```

Review

.git/
Main.java

```
package com.artusacademy

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

```
$ git init
Initialized empty Git repository in /app/.git/

$ git add .

$ git add .

$ git commit -m "Initial commit"
[master (root-commit) f228c1d] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 Main.java
```

untracked

staged

git repository

 "Hello World!"
Main.java

 f228c1d - Initial commit (HEAD → master) —  "Hello World!"
Main.java

Create another commit

- Add a integer variable `sum` whose value is `10 + 10`.
- Display the variable `sum` after `Hello World!`.
- Commit the code with message `Print sum`.

.git/
Main.java

```
package com.artusacademy

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    } System.out.println("Hello World!" + sum);
} }
```

```
$ git add .
$ git commit -m "Print sum"
[master 6c1735d] Print sum
1 file changed, 2 insertions(+), 1 deletion(-)
```

untracked

 "Hello World! 20"
Main.java

staged

git repository

6c1735d - Print sum
f228c1d - Initial commit (HEAD → master)

 "Hello World! 20"
Main.java

 "Hello World!"
Main.java

Connecting to remote

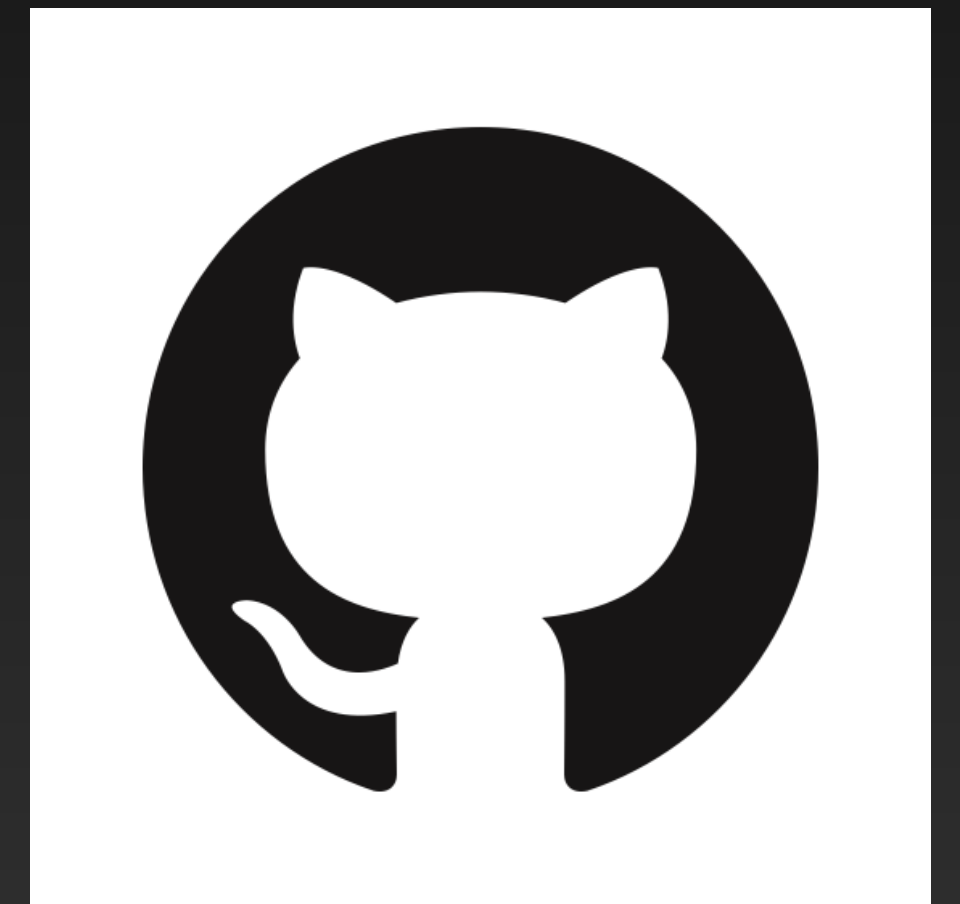
Commands

- `git remote`
- `git push`
- `git pull`

Github

<https://www.github.com>

- Largest web-based git hosting service
- Adds a UI for functionalities of git
- Founded in 2008



Create a GitHub account
(or login)

github.com

```
$ ssh-keygen -t ed25519 -f ~/.ssh/github
```

Generating public/private ed25519 key pair.

Enter passphrase (empty for no passphrase):

Enter same passphrase again:

Your identification has been saved in /users/home/john/.ssh/github

Your public key has been saved in /users/home/john/.ssh/github.pub

The key fingerprint is:

SHA256:43+iLVviM8DhrX4vIJqzJjjHVa3ZDQC+H7Yg3neAy1Y machine@localhost.localdomain

The key's randomart image is:

```
+--[ED25519 256]--+
|    ..           |
|    .  .         |
|    .  o         |
|    oo o        |
|  .  ++E*So      |
|  .  +o*B=o ..   |
| ... +=.+++  .   |
|+ B.  ..o*=. .   |
| =.o  ... +0+o   |
+---[SHA256]-----+
```

```
$ echo -e "Host github.com\n  IdentityFile ~/.ssh/git" > ~/.ssh/config
```

```
$ cat ~/.ssh/github.pub
```

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIF+XICi83Rwt2HGkWuwhRhDVjKKMocha2A8SpR7iqxdE machine@localhost.localdomain
```


Dashboard

Q

Type ↵ to search

nikhilnathr

Top repositories

New

Find a repository...

artus-academy/website

nikhilnathr/git-workshop

prodojo/workshops

nikhilnathr/flirterai

sidharth-n/flirterai

nikhilnathr/cryptonoteAPI-visualiser

nikhilnathr/alien-log-reader

Show more

Join GitHub Education!

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Copilot

Turn natural language prompts into coding suggestions

Heroku

Build, run, and operate applications entirely in the cloud.

Microsoft Azure

Access to Microsoft Azure cloud services and learning resources

Join GitHub Education

Latest from

Last week

Gemin

Visual

2 weeks

Reduc

runner

3 weeks

Improv

Action

3 weeks

Contro

organi

View cl

nikhilnathr

Nikhil Nath R

Set status

Profile

Repositories

Stars

Gists

Organizations

New task

Enterprises

Sponsors

Settings

Copilot settings

Feature preview

Appearance

Accessibility

Try Enterprise

Free

Sign out

Home

Ask anything

Go to settings page

Nikhil Nath (nikhilmnathr)
Your personal account [Switch settings context](#)

[Go to your personal profile](#)

- Public profile
- Account
- Appearance
- Accessibility
- Notifications

Access

- Billing and plans
- Emails
- Password and authentication
- Sessions
- SSH and GPG keys**
- Organizations
- Enterprises
- Moderation

SSH keys

This is a list of SSH keys associated with your account.

Authentication keys

**mac**
SHA256: QUIZIR/R7hQi5SCfTFMTYz9ywTeXVsa2smsFImnkuro
Added on Jul 31, 2024
Last used within the last 3 months — Read/write
[Delete](#)

Check out our guide to [connecting to GitHub using SSH keys](#) or troubleshoot [common SSH problems](#).

GPG keys

This is a list of GPG keys associated with your account. Remove any keys that you do not recognize.

**Email address:** nikhilmnath1000@gmail.com
Key ID: 9313C70D1E543DCE
[Delete](#)

Vigilant mode

- ☐ **Flag unsigned commits as unverified**
This will include any commit attributed to your account but not signed with your GPG or S/MIME key.
Note that this will include your existing unsigned commits.
[Learn about vigilant mode.](#)

1. Select “SSH and GPG keys”

2. Click “New SSH key” button

Add new SSH Key

1. Give a name to your key

Title

workshop-key

Key type

Authentication Key ▾

Key

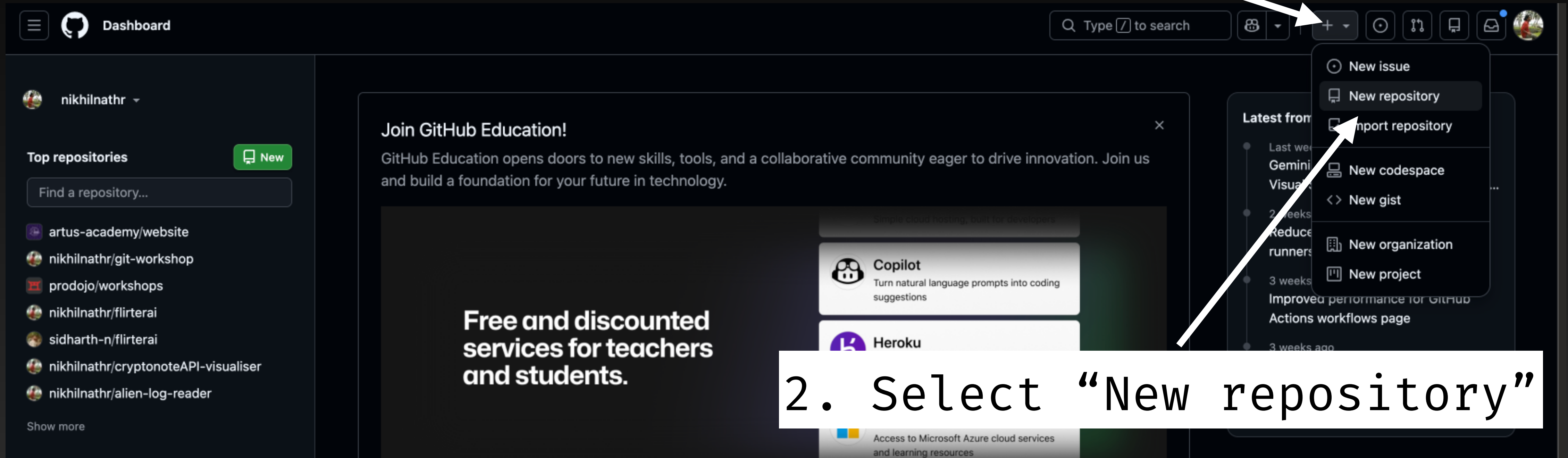
Begins with 'ssh-rsa', 'ecdsa-sha2-nistp256', 'ecdsa-sha2-nistp384', 'ecdsa-sha2-nistp521', 'ssh-ed25519', 'sk-ecdsa-sha2-nistp256@openssh.com', or 'sk-ssh-ed25519@openssh.com'

2. Paste you SSH public key

Add SSH key

3. Click "Add SSH key"

1. Click the “+” dropdown



The screenshot shows the GitHub dashboard for user 'nikhilmathr'. The top navigation bar includes a search bar and a '+ ' dropdown menu. The dropdown menu is open, showing options: 'New issue', 'New repository', 'Import repository', 'New codespace', 'New gist', 'New organization', and 'New project'. The 'New repository' option is highlighted. A white arrow points from the text '1. Click the “+” dropdown’ to the '+ ' button. Another white arrow points from the text '2. Select “New repository”’ to the 'New repository' option in the dropdown menu.

Dashboard

Q Type / to search

nikhilmathr

Top repositories

Find a repository...

artus-academy/website

nikhilmathr/git-workshop

prodojo/workshops

nikhilmathr/flirterai

sidharth-n/flirterai

nikhilmathr/cryptonoteAPI-visualiser

nikhilmathr/alien-log-reader

Show more

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Copilot

Turn natural language prompts into coding suggestions

Heroku

Access to Microsoft Azure cloud services and learning resources

Latest from

Last week

Gemini

Visual

2 weeks

Reduce

runners

3 weeks

Improved performance for GitHub Actions workflows page

3 weeks ago

New issue

New repository

Import repository

New codespace

New gist

New organization

New project

2. Select “New repository”

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1. Give a name to your repository

1 General

Owner *



Repository name *

artusacademy-workshop

✔ artusacademy-workshop is available.

Great repository names are short and memorable. How about [redesigned_robot?](#)

Description

A repository created as part of a workshop

43 / 350 characters

2. Select public (open-source ❤️)

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

3. Keep these options
unchecked
Or as "none"

Create repository

4. Create repository

Quick setup — if you've done this kind of thing before

 Set up in Desktop

or

HTTPS

SSH

git@github.com:nikhilnathr/artusacademy-workshop.git



Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

1. Select “SSH”

2. Copy this URL

git remote

<https://git-scm.com/docs/git-remote>

- Manage the set of repositories, also called **remotes**
- `git remote add origin <remote-url>`
- Here **origin** is the name of the remote, there can be other remotes also.


```
$ git remote add origin git@github.com:nikhilnathr/  
git-workshop.git  
$
```



git push

<https://git-scm.com/docs/git-push>

- Updates remote using local, by sending objects necessary to complete the given refs.
- `git push -u origin <branch_name>` - when pushing a branch for first time
- `git push` - for subsequent updates

```
$ git push -u origin master
```

```
Enumerating objects: 14, done.
```

```
Counting objects: 100% (14/14), done.
```

```
Delta compression using up to 8 threads
```

```
Compressing objects: 100% (11/11), done.
```

```
Writing objects: 100% (14/14), 1.43 KiB | 1.43 MiB/s, done.
```

```
Total 14 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
```

```
remote: Resolving deltas: 100% (1/1), done.
```

```
To github.com:johndoe/git-workshop.git
```

```
+ 8a4bdf6...1911cbb master → master
```

```
branch 'master' set up to track 'origin/master'.
```

```
$
```

repository

remotes

origin



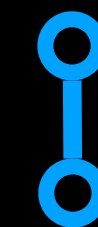
6c1735d - Print sum

master*

f228c1d - Initial commit



github.com - repository



6c1735d - Print sum

master

f228c1d - Initial commit

git pull

<https://git-scm.com/docs/git-pull>

- Incorporates changes from a remote repository into the current branch.

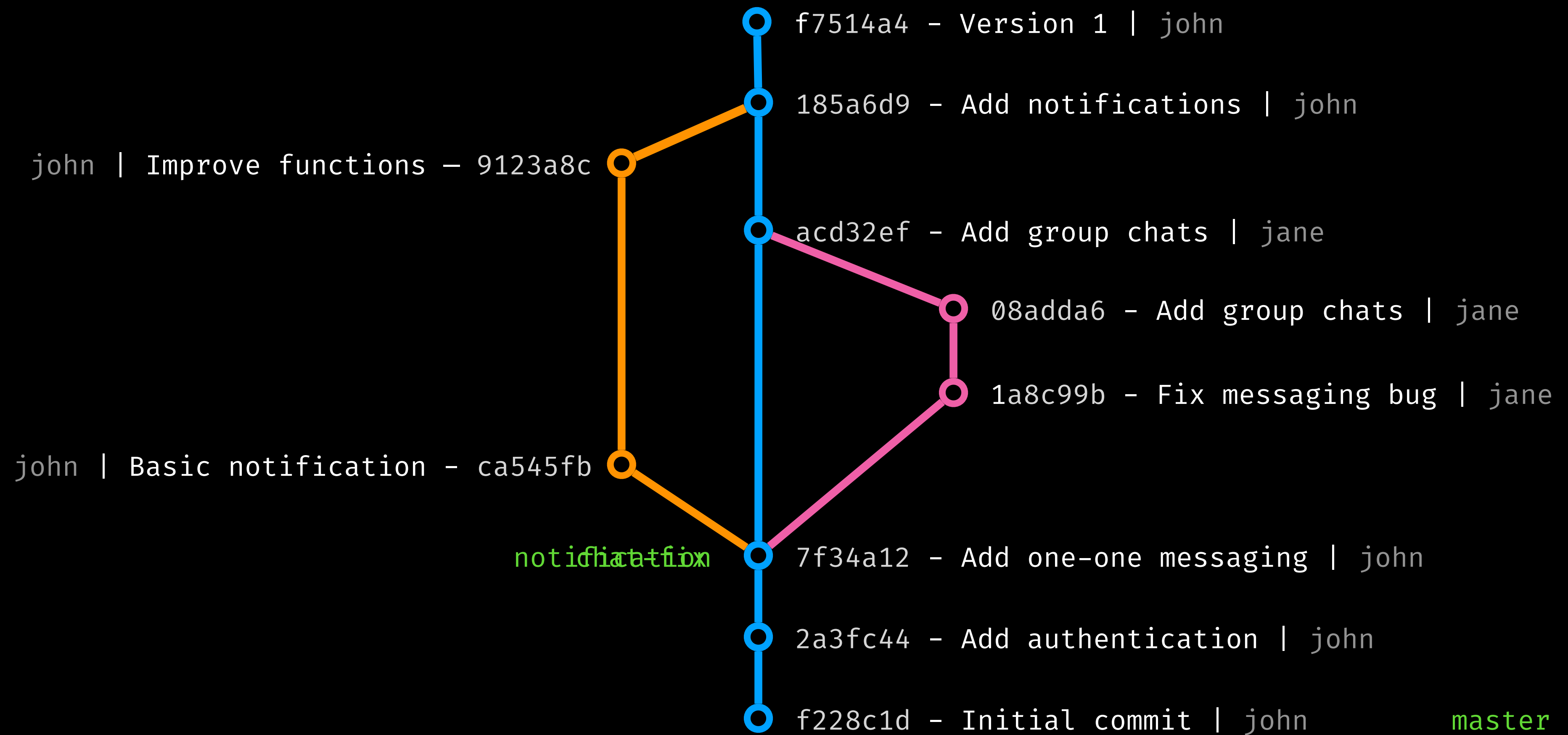
Branching

branches

- A bunch of commits linked together
- Branches can be worked from independently from each other
- The main branch is called **master** or **main** branch

merge

- The process of combining two separate branches into a single one
- Allows to combine the code or text from 2 separate features or developers



Commands

- `git branch`
- `git switch`
- `git merge`

git branch

<https://git-scm.com/docs/git-branch>

- Show the list of branches
- The current branch will be highlighted in green and marked with an asterisk.
- Show branches with,
`git branch`
- Create a branch with,
`git branch <branch_name>`

```
$ git branch
* master

$ git branch sum-function
$ git branch
    sum-function
* master

$
```

```
sum-function 6c1735d - Print sum master*
             f228c1d - Initial commit
```

git switch

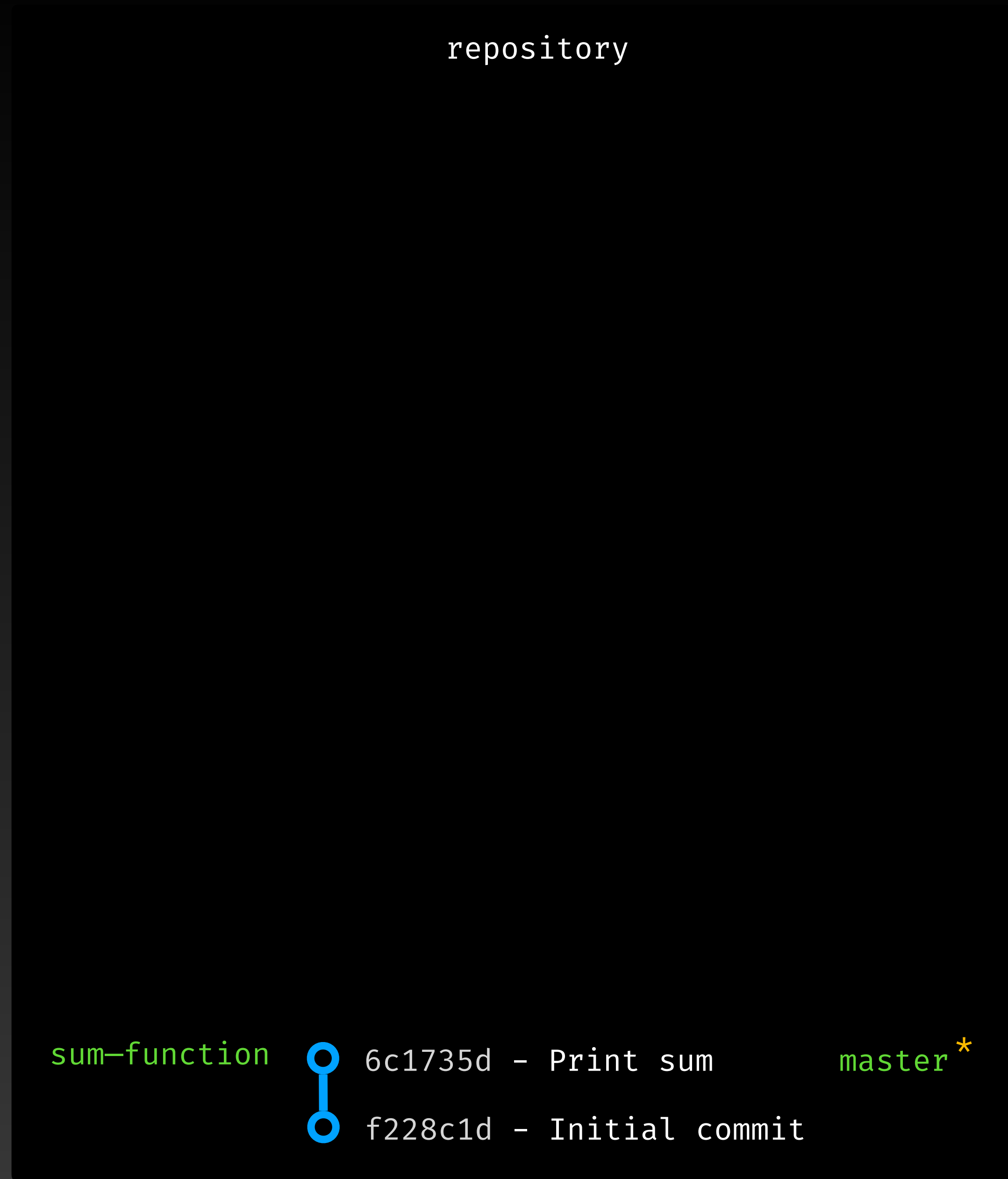
<https://git-scm.com/docs/git-switch>

- Switch to a specified branch.
- The working tree and the index are updated to match the branch.
- All new commits will be added to the tip of this branch.


```
$ git switch sum-function
Switched to branch 'sum-function'

$ git branch
* sum-function
  master

$
```



Commit in branch

- Add a **function add** in code for a function to calculate a **sum of two numbers**
- Use the function to calculate the sum.
- Commit using the message **“Add sum function”**

Main.java

```
public class Main {  
    public static void main(String[] args) {  
        int sum = add(10, 10);  
        System.out.println("Hello World");  
    }  
  
    private int add(int a, int b) {  
        return a + b;  
    }  
}
```

Do another commit

- Switch to `master` branch
- Create a file called `README.md`
- Commit with message “`Add README file`”

README.md

```
# Hello World
```

```
This is a hello world program written  
by nikhilnathr.
```


git merge

<https://git-scm.com/docs/git-merge>

- Join two or more development branches together
- To consolidate development work done in different branches.
- To do a merge
 - `git switch master`
 - `git merge <branch-name>`
 - (Or with message) `git merge <branch> -m "message"`

```
$ git merge sum-function -m "Merge sum function"
Merge made by the 'ort' strategy.
 hello.c | 6 +++++-
 1 file changed, 5 insertions(+), 1 deletion(-)

$
```

sum-function

Add sum function - adb41c8

repository

1911cbb - Merge sum function

ea7fb32 - Add README file master*

6c1735d - Print sum

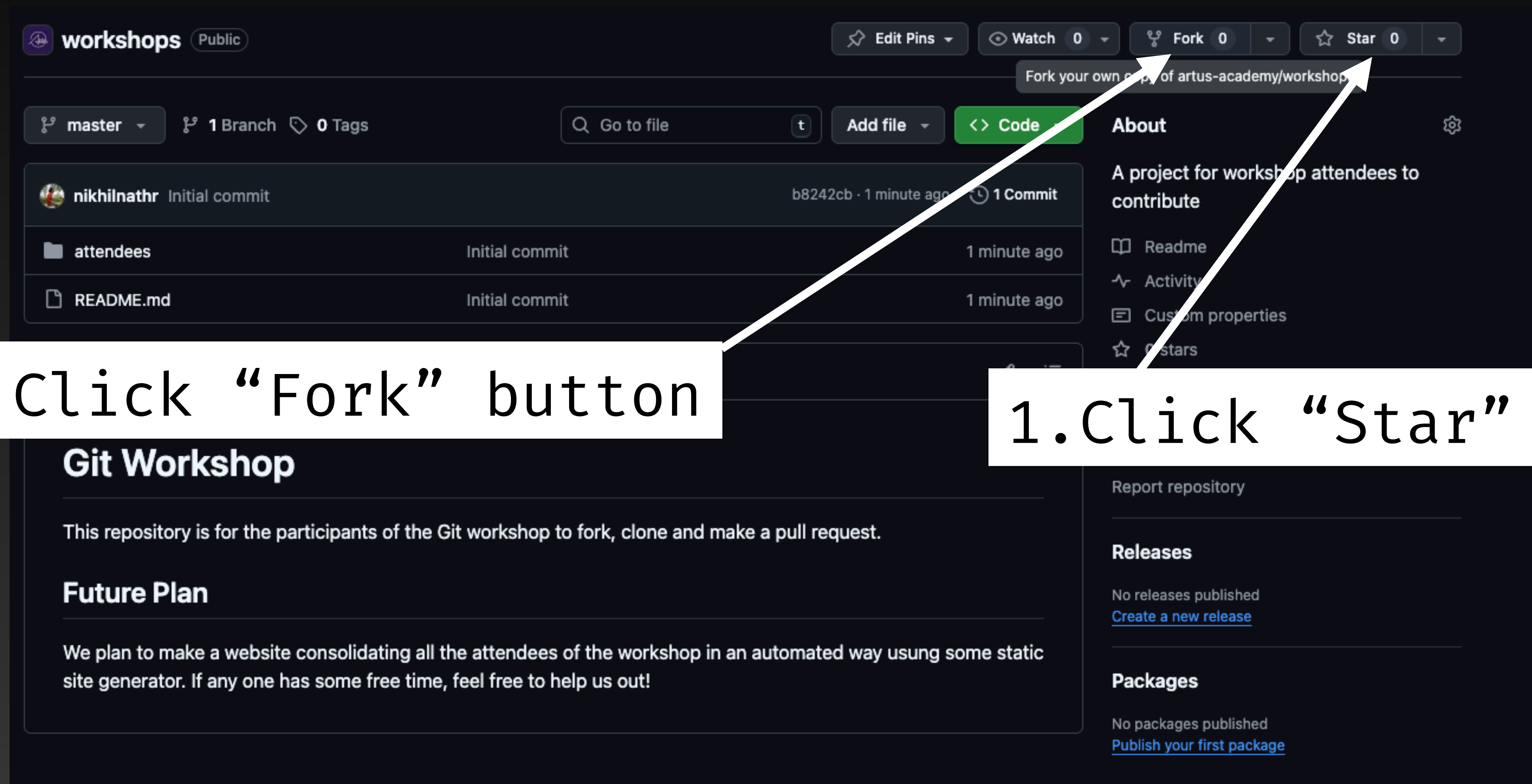
f228c1d - Initial commit

Contribute to project

forking

- A **fork** generally refers to creating a copy of a repository.
- This copy allows you to make changes independently from the original repository.
- On GitHub, when you fork a repository, it creates a personal copy of the repository under your own account.

<https://github.com/artus-academy/workshops>

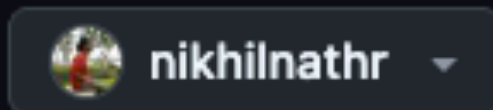


Create a new fork

A *fork* is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project.

Required fields are marked with an asterisk ().*

Owner *



Repository name *

workshops

✔ workshops is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

A project for workshop attendees to contribute

46 / 350 characters

☒ Copy the `master` branch only

Contribute back to artus-academy/workshops by adding your own branch. [Learn more.](#)

You are creating a fork in your personal account.

Create fork

Click “create fork” button

Commands

- `git clone`

git clone

<https://git-scm.com/docs/git-clone>

- Clones a repository into a newly created directory.
- Creates remote-tracking branches for each branch in the cloned repository

 **workshops** Public

[forked from artus-academy/workshops](#)

 master

 1 Branch

 0 Tags

Add file

 Code

This branch is up to date with `artus-academy/workshops:master`.

 **nikhilnathr** Initial commit

 attendees Initial commit

 README.md Initial commit

 README

Git Workshop

This repository is for the participants

Future Plan

We plan to make a website consolidating all the attendees of the workshop in an automated way using some static site generator. If any one has some free time, feel free to help us out!

LocalCodespaces

 Clone

HTTPSSSHGitHub CLI

`git@github.com:nikhilnathr/workshops.git`

Use a password-protected SSH key.

 Open with GitHub Desktop

 Download ZIP

1. Copy the repository link

workshops Public
forked from [artus-academy/workshops](#)

master 1 Branch 0 Tags

Go to file Add file Code

This branch is up to date with artus-academy/workshops:master .

nikhilnathr Initial commit

- attendees Initial commit
- README.md Initial commit

README

Git Workshop

1. Copy the repository link

Future Plan

We plan to make a website consolidating all the attendees of the workshop in an automated way using some static site generator. If any one has some free time, feel free to help us out!

```
$ git clone git@github.com:nikhilnathr/workshops.git
```

Cloning into 'workshops' ...

remote: Enumerating objects: 5, done.

remote: Counting objects: 100% (5/5), done.

remote: Compressing objects: 100% (4/4), done.

remote: Total 5 (delta 0), reused 5 (delta 0),
pack-reused 0 (from 0)

Receiving objects: 100% (5/5), done.

```
$ cd workshops
```

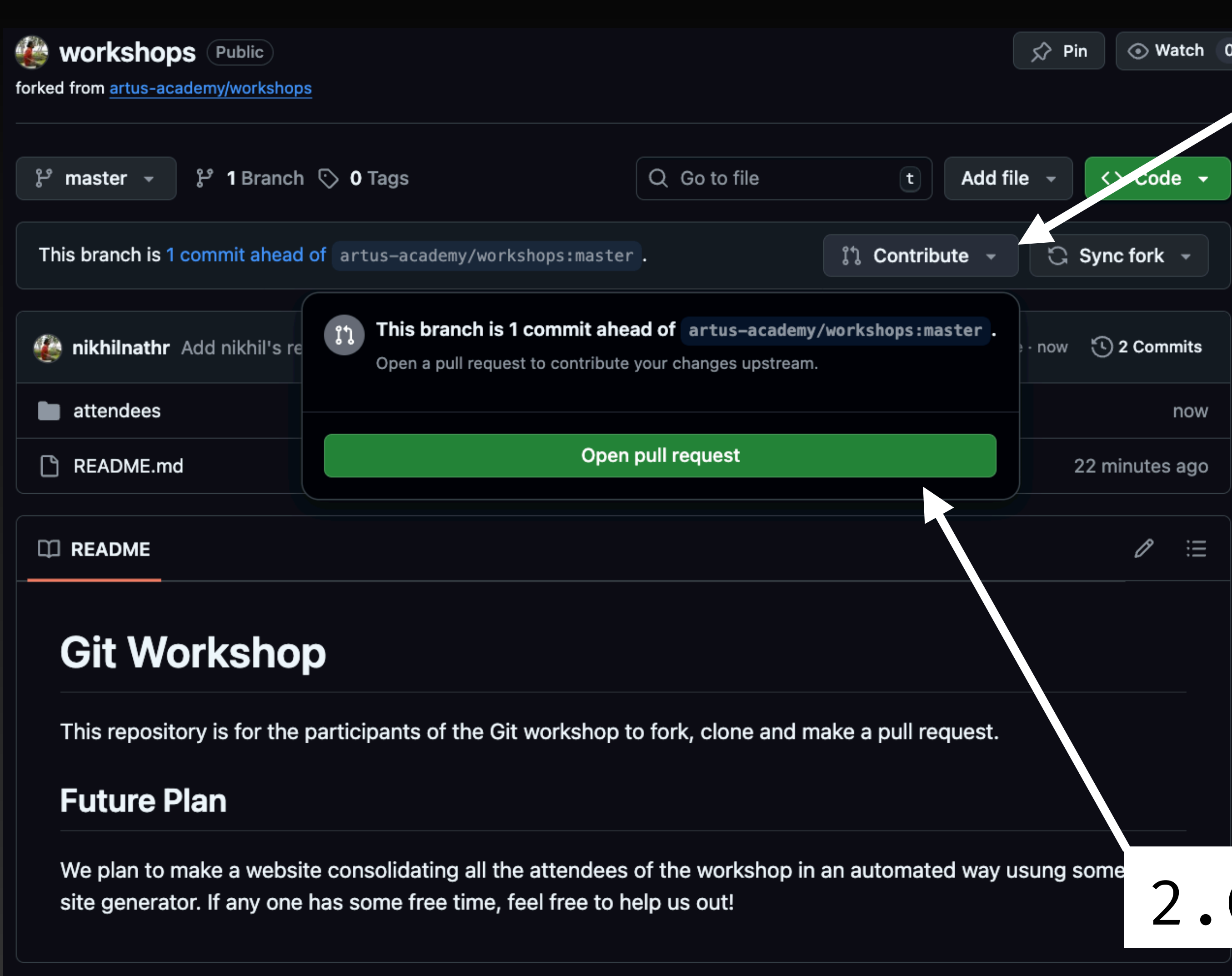
```
$
```


Task

- Go to `attendees` folder
- Copy `sample.md` to `<your_username>.md`
- Edit `<your_username>.md` with your details
- Make a commit
- Push to your repository

merge / pull request

- A pull request is a way to propose changes you've made in a branch or fork to the original repository.
- It allows collaborators to review, discuss, and suggest improvements on the proposed changes before merging them.



1. Click “contribute”

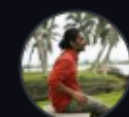
2. Click “Open pull request”

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more](#)

base repository: artus-academy/workshops base: master head repository: nikhilnathr/workshops compare: master

✓ **Able to merge.** These branches can be automatically merged.



Add a title

Add nikhilnathr as attendee

Add a description

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Add your description here...

Markdown is supported

Paste, drop, or click to add files

☒ Allow edits by maintainers ?

Create pull request

Reviewers

No reviews

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

No milestone

Development

Use [Closing keywords](#) in the description to automatically close issues

Helpful resources

[GitHub Community Guidelines](#)

Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

1. Add a title
“Add <username> as attendee”

2. Add a description if required

3. Click “Create pull request”

Clean the files

- `rm ~/.gitconfig` – remove git configuration file with your name and email
- `rm ~/.ssh/github.pub ~/.ssh/github` – remove your public key
- `rm ~/workshop` – Delete the project

Thank You

Nikhil Nath R

nikhil@artusacademy.com

+91 9946624506

artusacademy.com